

SigmaFlow

Riemannian Flow Matching for Generative Protein–Ligand Docking

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Abstract

Molecular docking predicts the bound geometry of a small molecule in a protein binding site. Recent generative approaches cast this as sampling from a conditional distribution over rigid-body poses, a space that is not Euclidean: a pose is an element of a product of special Euclidean groups. SigmaDock attacks this with score-based diffusion. This thesis replaces that generative mechanism with Riemannian flow matching while holding the network architecture, the data pipeline, and the computational budget fixed, so that the substitution itself — rather than a confounded collection of changes — is what is measured.

The resulting system, SigmaFlow, is compared to its diffusion parent at matched training compute. Early experiments reveal a sharp asymmetry between the two factors of the state space: the translational channel carries signal while the rotational channel is, at this budget, statistically indistinguishable from uniform guessing. We derive a candidate mechanism from the geometry of the construction. The geodesic conditional path on $SO(3)$ has a regression target that is *independent of time*, so flow matching demands the complete answer at $t = 0$, where it is not identifiable from the state; the L^2 -optimal response is then a conditional mean of rotation vectors, which contracts. Translation escapes this because its source distribution is informative and rotation's, being Haar, is not.

[RESULT PENDING] The abstract closes with the verdict of the matched-compute experiments and the source-distribution intervention. Rewrite the final two sentences once Chapter 7 is complete; the framing above is deliberately compatible with all four anticipated outcomes.

[CHECK WITH SUPERVISOR] Whether the abstract counts towards the 12,000-word limit, and whether appendices are excluded. The budget in this draft assumes both are excluded. This assumption materially changes Chapter 3's allocation and should be confirmed against the MSc Statistical Science handbook before the final build.

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Chapter 1

Introduction

1.1 Docking as a generative problem

Molecular docking asks where a small molecule — the *ligand* — sits when bound to a protein, and in what orientation and internal conformation. The answer determines whether a candidate compound is worth synthesising, and computational prediction has been a central tool of structure-based drug design for decades. Classical docking programs pose this as optimisation: define a scoring function that approximates binding free energy, then search the space of poses for its minimum. The weakness is well known — the scoring function is a crude surrogate, and the search returns a single point estimate with no calibrated notion of how much to trust it.

Generative modelling reframes the task. Instead of minimising a surrogate energy, one learns to *sample* from the conditional distribution $p(\text{pose} \mid \text{protein}, \text{ligand})$ implied by experimentally determined structures. Drawing K samples then yields a set of hypotheses whose spread is itself informative. DiffDock [4] established that this works at scale, and a line of subsequent work has refined the state space, the architecture and the noising process.

What makes the problem geometrically interesting is that a pose is not a vector. Placing a rigid body in space requires a rotation and a translation: an element of the special Euclidean group $\text{SE}(3)$. Treating a ligand as a small number of rigid fragments makes the configuration space a product $(\text{SE}(3))^N$, a compact-times-Euclidean manifold on which the usual machinery of Gaussian noise and linear interpolation does not directly apply. Generative models on such spaces must be built with the geometry, not retrofitted to it.

1.2 Research question and scope

SigmaDock [13] is a recent docking model that combines an $\text{SE}(3)$ -equivariant EquiformerV2 backbone [7] with a score-based diffusion process over rigid-fragment poses. It is the starting point of this thesis. Diffusion is not the only way to learn a transport between a tractable source distribution and the data; flow matching [8] learns a velocity field directly by regression, without simulating a stochastic process at training time, and its Riemannian extension [2] carries the construction to manifolds. The two approaches are closely related — diffusion admits

a deterministic probability-flow ODE [16] — but they differ in what the network is asked to predict, in how the source distribution is chosen, and in how many network evaluations sampling requires.

This thesis asks:

What changes when SigmaDock’s score-based diffusion is replaced by Riemannian flow matching on the protein–ligand pose manifold, while architecture, data and computational budget are held fixed — and which flow-specific design choices are required before the substitution becomes competitive?

The design principle throughout is *minimal intervention*. The backbone, featurisation, data pipeline, optimiser and evaluation protocol are inherited unchanged; only the generative mechanism is replaced. This is what makes the comparison interpretable, and Chapter 4 documents the extent to which it was achieved, including where it was not.

Two limitations are stated at the outset rather than buried. First, all experiments in this thesis are run at a fraction of the compute used to produce SigmaDock’s published numbers; absolute performance is therefore well below the literature and no benchmark claim is made. What is claimed is a *matched-budget* comparison between two generative engines. Second, the ligand is treated as a set of rigid fragments, so torsional flexibility enters only through the fragmentation and not as a modelled degree of freedom. Both constraints are inherited from SigmaDock and are shared by both arms of the comparison.

1.3 Contributions

1. A controlled replacement of the generative mechanism in a state-of-the-art docking system by Riemannian flow matching, with the network backbone verified byte-identical between arms (§4.1). The construction itself applies known theory; the *control* is what makes the comparison usable.
2. Identification and correction of two defects in the rotational machinery: a frame inconsistency in the predicted vector field (§4.2) and a clamping error in the logarithmic map inherited from the parent codebase (§4.3). Both are documented with the behavioural signature of the fix.
3. A derivation showing that the geodesic conditional target on $\text{SO}(3)$ is independent of time, and the resulting shrinkage argument for why this removes the implicit curriculum a diffusion noise schedule provides (Chapter 5). This is the principal theoretical contribution.
4. An empirical programme testing that mechanism against training compute, together with the exclusion of three competing explanations — integration resolution, molecular symmetry and loss weighting (§7.5.2).
5. A leakage-controlled construction for an informative source distribution on $\text{SO}(3)$, motivated directly by the mechanism rather than imported from elsewhere (§7.6).

[UPDATE AFTER FINAL RUNS] Re-rank this list once Chapters 7 and 8 are complete. If the informative-source experiment succeeds, item 5 moves up; if the rotational deficit persists under longer training, item 3 becomes the headline.

Chapter 2

Docking as Conditional Generation on a Product Manifold

2.1 Poses, rigid fragments, and the state space

Let a ligand consist of N_a atoms. A *pose* is an assignment of positions to those atoms consistent with the molecule’s covalent geometry. Modelling the full conformational space couples bond lengths, angles and torsions; SigmaDock instead partitions the ligand into N rigid *fragments*, treats internal fragment geometry as fixed at a precomputed conformer, and generates only the rigid-body placement of each fragment. Torsional flexibility survives implicitly, as the relative placement of neighbouring fragments, but is not itself a modelled coordinate.

A single fragment’s placement is an element of the special Euclidean group,

$$\text{SE}(3) = \{ (R, x) : R \in \text{SO}(3), x \in \mathbb{R}^3 \}, \quad (2.1)$$

acting on a body-frame atom offset y by $y \mapsto Ry + x$. The full state is therefore a point on the product manifold

$$\mathcal{M} = (\text{SE}(3))^N \cong (\text{SO}(3))^N \times (\mathbb{R}^3)^N, \quad \dim \mathcal{M} = 6N. \quad (2.2)$$

Two remarks matter for what follows. First, $\mathcal{M}$ is a *product* of a compact factor and a Euclidean one, and the implementation treats it as a direct product: rotations and translations are given independent probability paths and independent vector fields, with no cross terms. This is a modelling choice, not a property of $\text{SE}(3)$, and it is stated explicitly here because the thesis’s central finding is precisely an asymmetry *between* the two factors. Second, N is small: over the 209 evaluation complexes the median fragment count gives $\dim \mathcal{M} = 24$, with a ninetieth percentile of 42 and a maximum of 66. The state space is low-dimensional; the difficulty lies in its geometry and in the conditioning, not in dimension.

Translations are expressed relative to the binding-pocket centroid, so the origin of the $\mathbb{R}^3$ factor is a chemically meaningful point. This apparently minor convention becomes important in Chapter 5.

2.2 The conditional generative formulation

Write c for the conditioning context: the protein pocket, the ligand’s molecular graph, and the precomputed fragment conformers. The learning target is the conditional law of the crystallographic pose,

$$p_{\text{data}}(\cdot | c) \quad \text{on} \quad \mathcal{M}, \quad (2.3)$$

from which we wish to draw samples at inference. Two features distinguish this from unconditional generative modelling on a manifold. The conditioning is strong — given a pocket, the plausible poses occupy a small region — and the training data provide exactly *one* sample per context, namely the deposited structure. There is no set of ground-truth poses per complex to match in distribution, only a single point. Every generative construction below must therefore be conditional on c and trained from single observations.

2.3 SigmaDock: the score-based baseline

SigmaDock defines a forward noising process that carries the data pose towards a tractable reference: an isotropic Gaussian on each translational factor and the heat kernel on each rotational factor, whose density is the IGSO(3) distribution obtained by solving the diffusion equation on $\text{SO}(3)$ [6, 18]. A network is trained by denoising score matching [15, 17] to estimate the Stein score $\nabla \log p_\tau$ of the perturbed marginals at noise level τ , and sampling integrates the reverse-time stochastic differential equation [16]. Poses are generated in batches of K and ranked by a fixed heuristic combining a Vinardo binding energy [14] with a stereochemical validity fraction; the published headline numbers are post-ranking.

2.3.1 The inherited backbone, and how much of it the reader needs

The network is an EquiformerV2 [7], and it is inherited by SigmaFlow *without modification*. Rather than reproduce its internals, this section states only the interface, which is what the argument of this thesis actually requires. Appendix A gives the representation-theoretic background for readers who want it.

Physical correctness demands that the model be equivariant: rotating the entire complex must rotate the predicted pose update identically, and permuting atom indices must permute the outputs. A network built purely from distances and other invariant scalars satisfies the first requirement trivially but cannot then *emit* a vector, since any function of invariants is itself invariant. The resolution is to carry features that transform in a controlled way — so-called steerable features, decomposed into components indexed by degree ℓ , where $\ell = 0$ components are scalars invariant under rotation and $\ell = 1$ components transform as vectors. Message passing is then designed so that this transformation law is preserved through every layer.

The consequence for this thesis is narrow and concrete. The backbone maps the conditioning context and the current state to a per-atom steerable feature tensor. A single output block

reads from it and emits *one* $\ell = 1$ vector per ligand atom,

$$f_i \in \mathbb{R}^3, \quad i = 1, \dots, N_a, \quad (2.4)$$

which we refer to as the force field by analogy with rigid-body mechanics. Both rigid-body degrees of freedom of a fragment F are then obtained from this single field by the Newton–Euler reduction

$$\text{translation: } F_{\text{net}} = \sum_{i \in F} f_i, \quad (2.5)$$

$$\text{rotation: } \tau = \sum_{i \in F} (r_i - c_F) \times f_i, \quad (2.6)$$

where c_F is the fragment centroid and r_i the atom position. There is no separate rotational head; the two channels are two *linear functionals of the same output*. Since $\sum_{i \in F} (r_i - c_F) = 0$, a spatially constant component of f produces net force but zero torque, and a purely rotational pattern produces torque but zero force: the functionals are orthogonal. Chapter 5 and §7.3 return to the very different *conditioning* of these two readouts.

FIGURE PLACEHOLDER

Schematic in three columns. **Left (inherited, grey):** protein pocket and ligand graph $\rightarrow$ featurisation $\rightarrow$ EquiformerV2 backbone $\rightarrow$ per-atom steerable features. **Middle (inherited, grey):** single $\ell = 1$ output block emitting $f_i \in \mathbb{R}^3$ per atom; Newton–Euler reduction to (F_{net}, τ) per fragment, Eqs. (2.4)–(2.5). **Right (replaced, blue):** interpretation layer. SigmaDock reads (F_{net}, τ) as a *score* and integrates a reverse SDE; SigmaFlow reads them as a *velocity* and integrates an ODE. Annotate the right column with the two objectives, Eq. (3.20) and the diffusion counterpart. The figure must make visually obvious that the substitution is confined to one block — this is the evidence for the controlled comparison and should replace roughly 400 words of prose.

Figure 2.1: Inherited architecture and the interface to the generative mechanism. Components shaded grey are byte-identical between SigmaDock and SigmaFlow; only the shaded-blue block is replaced.

2.4 Evaluation and the random-rotation null

Docking accuracy is reported as the root-mean-square deviation (RMSD) between predicted and crystallographic ligand atoms, with the fraction of complexes below 2 Å the conventional success criterion. Because RMSD mixes placement error with internal geometry error, we also report RMSD after optimal rigid superposition [5], which isolates internal geometry, and the centroid displacement, which isolates placement. Physical plausibility is assessed with the PoseBusters battery [1].

For the rotational channel a null model is essential. If fragment orientations were drawn uniformly (Haar) on $\text{SO}(3)$, the expected geodesic angle to the truth would be

$$\mathbb{E}[\theta] = \frac{\pi^2/2 + 2}{\pi} \approx 126.5^\circ, \quad (2.7)$$

with a median of about 132.3° under the sampling protocol used here. Every rotational number

in Chapter 7 is reported against this reference, because a rotation error of 130° is not a poor prediction — it is no prediction at all.

Chapter 3

From Score to Velocity: Riemannian Flow Matching

This chapter develops the generative machinery that replaces diffusion in SigmaFlow. The treatment is selective: results that are standard and not load-bearing for the argument are stated and cited, while two developments — the conditional objective and the geodesic construction on $\text{SO}(3)$ — are derived, because Chapters 5 and 7 depend on their precise form.

3.1 Continuous normalising flows and the intractable marginal field

Let $\mathcal{M}$ be the state space and let u_t be a time-dependent vector field on it. The associated flow ψ_t solves

$$\frac{d}{dt}\psi_t(x) = u_t(\psi_t(x)), \quad \psi_0(x) = x. \quad (3.1)$$

Pushing a source distribution p_0 through ψ_t produces a path of distributions $p_t = (\psi_t)_\# p_0$, and this is a continuous normalising flow [3]. Generative modelling reduces to finding a field whose flow transports a tractable p_0 to the data distribution p_1 .

If such a field is known, sampling is straightforward: draw $x_0 \sim p_0$ and integrate (3.1). The difficulty is training. A field u_t generates p_t precisely when the pair satisfies the continuity equation, $\partial_t p_t + \nabla \cdot (p_t u_t) = 0$, and the natural objective

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t, x \sim p_t} \|v_\theta(t, x) - u_t(x)\|^2 \quad (3.2)$$

cannot be evaluated, because neither p_t nor u_t is available in closed form for any interesting p_1 . Flow matching resolves this [8].

3.2 Conditional flow matching

The construction conditions on the data point. For each x_1 , choose a conditional path $p_t(\cdot | x_1)$ with $p_0(\cdot | x_1) = p_0$ and $p_1(\cdot | x_1) = \delta_{x_1}$, generated by a conditional field $u_t(\cdot | x_1)$ that we may write down by hand. The marginal path and its field are then

$$p_t(x) = \int p_t(x | x_1) q(x_1) dx_1, \quad u_t(x) = \int u_t(x | x_1) \frac{p_t(x | x_1) q(x_1)}{p_t(x)} dx_1, \quad (3.3)$$

where q is the data distribution. The second expression is exactly a conditional expectation,

$$u_t(x) = \mathbb{E}[u_t(x | X_1) | X_t = x]. \quad (3.4)$$

This identity is the pivot of the whole method and is used again, for a quite different purpose, in Chapter 5.

Proposition 3.1 (Marginalisation). *Define the conditional objective*

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t, x_1 \sim q, x \sim p_t(\cdot | x_1)} \|v_\theta(t, x) - u_t(x | x_1)\|^2. \quad (3.5)$$

Then $\nabla_\theta \mathcal{L}_{\text{CFM}} = \nabla_\theta \mathcal{L}_{\text{FM}}$, so the two objectives have the same minimisers in θ .

Proof sketch. Expand both squared norms. The terms $\|v_\theta\|^2$ agree because both expectations are over the same marginal p_t . The terms not involving θ do not affect the gradient. The cross terms agree because, by (3.4) and the tower property, $\mathbb{E}_{p_t} \langle v_\theta(t, x), u_t(x) \rangle = \mathbb{E}_{p_t} \langle v_\theta(t, x), \mathbb{E}[u_t(x | X_1) | X_t = x] \rangle = \mathbb{E}_{x_1, x} \langle v_\theta(t, x), u_t(x | x_1) \rangle$. $\square$

Two consequences follow, and they pull in different directions.

The practical one is that training becomes a regression onto a target written down in closed form. Nothing is simulated: one samples t , samples a data point, samples a point on its conditional path, evaluates $u_t(x | x_1)$ analytically, and takes a gradient step. Diffusion models achieve something similar through denoising score matching [17], but there the tractable target arises from a Gaussian transition kernel, whereas here the path is chosen first and the target follows from it.

The second consequence is subtler and is the one this thesis turns on. The minimiser of (3.5) is not the conditional field — that object depends on x_1 , which the network never sees — but its conditional expectation (3.4) given the current state. This is exactly the standard fact that an L^2 regression returns a conditional mean. Its force depends entirely on how much X_t reveals about X_1 . If the state nearly determines the data point, the conditional mean is close to a single target and the regression is well posed. If it does not, the model is trained towards an average of many incompatible targets, and that average may be a poor member of the set it averages. Chapter 5 shows that the two factors of the docking state space sit at opposite ends of this spectrum.

In $\mathbb{R}^d$ the standard choice is the straight-line path

$$x_t = (1 - t)x_0 + tx_1, \quad u_t(x_t \mid x_0, x_1) = x_1 - x_0, \quad (3.6)$$

whose conditional trajectories are straight and whose target is constant along each one. This is what SigmaFlow uses on the translational factor.

3.3 The Riemannian generalisation

On a Riemannian manifold $(\mathcal{M}, g)$ several ingredients of the Euclidean construction stop making sense, and it is worth being precise about which.

The interpolant $(1 - t)x_0 + tx_1$ is not defined, because there is no addition. The difference $x_1 - x_0$ is not defined either, and neither is the statement that a velocity “is” a point of the space: a velocity at x is an element of the tangent space $T_x\mathcal{M}$, and tangent spaces at different points are different vector spaces that cannot be compared without extra structure. Finally, the squared error $\|v_\theta - u\|^2$ must be measured in the metric at the point where both vectors live.

What survives is the part that matters. Proposition 3.1 rests only on bilinearity of an inner product and the tower property of conditional expectation, both of which hold pointwise on a manifold, so the conditional objective remains a valid surrogate [2]. Three substitutions then carry the construction over: straight lines become geodesics; differences become logarithmic maps; and the Euclidean norm in (3.5) becomes the Riemannian metric at the relevant point,

$$\mathcal{L}_{\text{RCFM}}(\theta) = \mathbb{E}_{t, x_1, x_t} \|v_\theta(t, x_t) - u_t(x_t \mid x_1)\|_{g(x_t)}^2. \quad (3.7)$$

The geodesic interpolant between x_0 and x_1 is

$$x_t = \text{Exp}_{x_0}(t \text{Log}_{x_0}(x_1)), \quad (3.8)$$

where $\text{Exp}_x : T_x\mathcal{M} \rightarrow \mathcal{M}$ follows the geodesic leaving x with a given initial velocity and Log_x is its local inverse. Differentiating (3.8) and re-expressing the result at x_t gives the conditional field

$$u_t(x_t \mid x_1) = \frac{\text{Log}_{x_t}(x_1)}{1 - t}, \quad (3.9)$$

which is the statement that at time t one must travel the remaining geodesic distance in the remaining time. Equation (3.9) is the object whose behaviour on $\text{SO}(3)$ drives Chapter 5.

3.4 The rotation group

$\mathrm{SO}(3) = \{R \in \mathbb{R}^{3 \times 3} : R^\top R = I, \det R = 1\}$ is a compact three-dimensional Lie group. Its Lie algebra $\mathfrak{so}(3)$ is the space of skew-symmetric matrices, identified with $\mathbb{R}^3$ by the hat map

$$\hat{\omega} = \begin{pmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{pmatrix}, \quad \omega \in \mathbb{R}^3. \quad (3.10)$$

With the bi-invariant metric $\langle A, B \rangle = \frac{1}{2} \mathrm{tr}(A^\top B)$ the exponential map is Rodrigues' formula and its inverse is available in closed form for $\theta \in (0, \pi)$:

$$\mathrm{Exp}(\hat{\omega}) = I + \frac{\sin \theta}{\theta} \hat{\omega} + \frac{1 - \cos \theta}{\theta^2} \hat{\omega}^2, \quad \theta = \|\omega\|, \quad (3.11)$$

$$\mathrm{Log}(R) = \frac{\theta}{2 \sin \theta} (R - R^\top), \quad \theta = \arccos\left(\frac{\mathrm{tr} R - 1}{2}\right). \quad (3.12)$$

The choice of a *bi-invariant* metric is not cosmetic. It makes left and right translation isometries, so distances are unchanged by re-expressing both arguments in another frame; it makes the geodesics through the identity exactly the one-parameter subgroups $t \mapsto \mathrm{Exp}(t\hat{\omega})$, which is what puts (3.11) in closed form; and it makes the Haar measure the Riemannian volume, so “uniform on $\mathrm{SO}(3)$ ” is unambiguous. Every closed form in this chapter depends on it, and it is available because $\mathrm{SO}(3)$ is compact.

The geodesic distance between R_a and R_b is $\|\mathrm{Log}(R_a^\top R_b)\|$, taking values in $[0, \pi]$. The logarithm is branch-ambiguous at $\theta = \pi$, the cut locus: a rotation by exactly π about an axis n is equally a rotation by $-\pi$ about the same axis, and Log must choose. This is not a rare edge case here. Under the Haar measure the angle has density proportional to $1 - \cos \theta$, which is *increasing* on $[0, \pi]$, so uniformly sampled rotations concentrate towards large angles: roughly one draw in ten exceeds 170° . Since the rotational source of §3.5 is exactly Haar, a tenth of all training pairs begin near the cut locus, and the numerical treatment of (3.12) there is a correctness requirement rather than a refinement. §4.3 reports a defect of exactly this kind.

3.4.1 Trivialisation, fixed explicitly

Because $\mathrm{SO}(3)$ is a group, a tangent vector at R may be transported to the identity in two inequivalent ways, and the resulting conventions are easy to conflate. Writing a curve's velocity as

$$\dot{R} = R \hat{\Omega} \iff \hat{\Omega} = R^\top \dot{R} \quad (\text{body frame; left-trivialised}) \quad (3.13)$$

differs from

$$\dot{R} = \hat{w} R \iff \hat{w} = \dot{R} R^\top \quad (\text{spatial frame; right-trivialised}), \quad (3.14)$$

and the two are related by the adjoint action,

$$\widehat{\Omega} = R^\top \widehat{w} R = \text{Ad}_{R^{-1}} \widehat{w}. \quad (3.15)$$

This thesis uses the *body-frame* convention (3.13) throughout, following Marsden and Ratiu [11]. Every rotational velocity, target and network output is a body-frame quantity, and the integrator correspondingly right-multiplies. §4.2 documents a defect that arose from mixing (3.13) and (3.14), and the terminological trap that produced it.

3.5 The SigmaFlow probability path and objective

Specialising (3.8) and (3.9) to the two factors of $\mathcal{M}$ gives the construction implemented in SigmaFlow. On each translational factor, with $x_0 \sim \mathcal{N}(0, I_3)$ in pocket-centred coordinates,

$$x_t = (1 - t)x_0 + t x_1, \quad u_t^{\text{trans}} = x_1 - x_0 = \frac{x_1 - x_0}{1 - t}. \quad (3.16)$$

On each rotational factor, with $R_0 \sim \mathcal{U}_{\text{SO}(3)}$ drawn from the Haar measure,

$$R_t = R_0 \text{Exp}(t \text{Log}(R_0^\top R_1)), \quad u_t^{\text{rot}} = \frac{\text{Log}(R_t^\top R_1)}{1 - t}. \quad (3.17)$$

Both expressions in (3.16) are equal, and the corresponding identity for (3.17) is proved in Chapter 5; the implementation uses whichever form is better conditioned. The right-hand forms, involving x_t and R_t , express the target as “the remaining distance divided by the remaining time” and are singular at $t = 1$; the left-hand forms, involving only the endpoints, are not.

The product structure of $\mathcal{M}$ makes the extension to N fragments immediate. On a Riemannian product the metric is the direct sum of the factor metrics, geodesics are products of factor geodesics, and Exp and Log act componentwise. A path constructed independently on each factor is therefore a legitimate path on $\mathcal{M}$, and the conditional field is the concatenation of the factor fields. Since the factors are treated as independent, the objective is a weighted sum of the two factor-wise losses,

$$\mathcal{L}(\theta) = \mathbb{E}_{t \sim \mathcal{U}[0,1]} \left[\lambda_{\text{trans}} \|v_\theta^{\text{trans}} - u_t^{\text{trans}}\|^2 + \lambda_{\text{rot}} \|v_\theta^{\text{rot}} - u_t^{\text{rot}}\|^2 \right], \quad (3.18)$$

with the weights inherited unchanged from SigmaDock’s production configuration. Sampling integrates the learned field by explicit Euler, using the exponential map as the retraction on the rotational factor:

$$x^{(k+1)} = x^{(k)} + v_\theta^{\text{trans}} \Delta t, \quad R^{(k+1)} = R^{(k)} \text{Exp}(v_\theta^{\text{rot}} \Delta t). \quad (3.19)$$

The right-multiplication in (3.19) is the concrete expression of the body-frame convention fixed in §3.4.1.

3.6 Score versus velocity

The substitution changes what the network means, not what it outputs. Both models emit the per-atom field (2.4) and reduce it by (2.5)–(2.6). SigmaDock interprets the result as a score $\nabla \log p_\tau$, whose magnitude must be rescaled by noise-level-dependent factors before it can be used in an update; SigmaFlow interprets it as a velocity, which is already in the units of the state and requires no rescaling. The entire scalar apparatus that converts scores into displacements is therefore deleted rather than modified (§4.1).

The two formulations are relatives. A diffusion model admits a deterministic probability-flow ODE whose velocity is an affine function of the score [16], so diffusion may be viewed as flow matching along a particular, Gaussian, probability path [8]. What distinguishes them in practice is that flow matching makes the path an explicit design choice, and that sampling is deterministic.

3.7 The source distribution as a modelling choice

One consequence of that freedom deserves emphasis, because Chapter 5 turns on it and §7.6 exploits it. Under diffusion the source is not chosen: it is whatever the forward noising process converges to, an isotropic Gaussian on $\mathbb{R}^3$ and the uniform measure on $\text{SO}(3)$. Under flow matching the only requirements on p_0 are that it be tractable to sample and that the conditional path connect it to p_1 . Any such p_0 is admissible, including one that depends on the conditioning context c .

This is a genuinely flow-specific degree of freedom, and it is not merely a convenience. The quality of the learning problem depends on how much the source already knows. SigmaFlow’s translational source is centred on the binding pocket and is therefore informative; its rotational source is Haar and carries no orientational information whatsoever. Chapter 5 argues that this difference has consequences that the geometry makes precise.

Chapter 4

SigmaFlow: The Minimal Conversion

4.1 Design principle and what actually changed

The comparison in Chapter 7 is only interpretable if the substitution is confined to the generative mechanism. This was treated as a constraint on the implementation rather than an aspiration, and the resulting partition is given in Table 4.1.

Table 4.1: Partition of the system under the minimal-change constraint. “Identical” was verified by file hash over the relevant source trees.

Component	Status	Note
EquiformerV2 backbone	identical	all backbone source files hash-identical between arms
Featurisation, graph construction	identical	inherited
Data pipeline, fragmentation	identical	same conformers, same splits
Output block ($\ell = 1$ head)	identical	one field f_i ; Eq. (2.4)
Newton–Euler reduction	identical	Eqs. (2.5)–(2.6)
Optimiser, LR schedule	identical	AdamW [10], same cosine schedule
Loss weights	identical	SigmaDock production values
Forward process	removed	no noising process; replaced by a probability path
Score interpretation and rescaling	removed	the scalar apparatus converting scores to displacements is deleted, not modified
Probability path	new	Eqs. (3.16)–(3.17)
Training target	new	conditional velocity in place of score
Sampler	new	deterministic Euler, Eq. (3.19), in place of reverse SDE
Logarithmic map clamp	corrected	§4.3
Rotational frame convention	corrected	§4.2

Two residual confounders should be recorded rather than glossed. The two arms run in separate software environments with different interpreter versions, and the SigmaDock arm is invoked with its own rotational-score configuration flags. Neither touches the backbone, but neither is strictly nothing, and §8.3 returns to them.

4.2 A frame defect in the rotational field

The convention fixed in §3.4.1 requires that the network’s rotational output be a *body-frame* quantity, since the integrator right-multiplies by its exponential. The Newton–Euler reduction (2.6), however, computes a torque from atom positions expressed in the global frame, and therefore returns a spatial-frame vector. In the original implementation this spatial quantity was passed directly to the integrator, mixing (3.13) with (3.14).

The correction is the adjoint transport (3.15): the predicted rotational velocity must be conjugated into the body frame,

$$\hat{\Omega}_{\text{body}} = R_t^\top \hat{w}_{\text{spatial}} R_t, \quad (4.1)$$

before it enters $\text{Exp}(\cdot \Delta t)$. Two facts make this worth a section rather than a footnote. First, the error is invisible to any check that does not vary R_t : at $R_t = I$ the two frames agree exactly, so a unit test at the identity passes. Second, the error is not small — taking an Euler step of length $1 - t$ under the correct convention reaches R_1 to within 5×10^{-4} degrees, whereas the incorrect left-multiplication misses by roughly 179.5° .

A terminological hazard contributed to the defect and is worth stating, because the literature is not consistent. The implementation’s own comments describe the construction as “right-trivialised” while implementing $\dot{R} = R\hat{\Omega}$, which is the *left*-trivialised or body convention in the sense of Marsden and Ratiu [11]. The mathematics was correct; the label was not. §7.5 reports the behavioural signature of the correction, and Appendix D the numerical verification.

4.3 A clamping defect in the logarithmic map

The inherited implementation of (3.12) clamped the argument of arccos to $[-0.99, 0.99]$, a guard intended to avoid a domain error. The effect is to misreport every angle above roughly 172° — close to one in ten draws from the Haar measure, by the cut-locus remark in §3.4.

For diffusion this is harmless, because the reverse SDE never requires an exact round trip $\text{Exp}(\text{Log}(R)) = R$. For the geodesic construction (3.17) it is not, because the interpolant is defined by exactly that round trip: a misreported Log places R_t off the intended geodesic and corrupts the regression target for a non-negligible fraction of training pairs. The clamp was narrowed to $[-1 + 10^{-7}, 1 - 10^{-7}]$.

This is an instance of a general point worth making once: numerical guards inherited from one generative formulation are not automatically valid in another, because the two formulations exercise the geometry differently.

4.4 Deterministic integration

Sampling integrates (3.19) from $t = 0$ to $t = 1$ over a fixed number of steps. On the translational factor this is ordinary explicit Euler; on the rotational factor the update is a retraction, meaning

the step is taken along the manifold rather than in an ambient space followed by a projection. Because Exp maps into $\text{SO}(3)$ exactly, every iterate is a rotation matrix by construction and no re-orthogonalisation is required — a small but real advantage of working in the group rather than with a matrix parameterisation.

Three properties distinguish this from the reverse-time SDE it replaces. Sampling is *deterministic*, so the pose is a function of the source draw alone and the diversity of a K -sample set is entirely inherited from p_0 ; the sampler has no temperature or noise-scale to tune. The number of network evaluations equals the number of steps and is a free parameter at inference, decoupled from anything fixed at training time. And because the conditional paths are straight in $\mathbb{R}^3$ and geodesic on $\text{SO}(3)$, the marginal field they induce is expected to be nearly straight, so few Euler steps should suffice — a prediction of the construction rather than a hope. §7.4 tests it over a forty-fold range of step counts.

The step count inherited from SigmaDock is 25, a value tuned there for a *stochastic* reverse process. There is no reason for that number to transfer, and §7.4 shows that it does not.

FIGURE PLACEHOLDER

Two panels. **Left:** $\mathbb{R}^3$ (drawn as $\mathbb{R}^2$). Source sample x_0 from a Gaussian centred at the pocket origin; data point x_1 ; the straight interpolant with the constant target $x_1 - x_0$ drawn as equal arrows at $t = 0.2, 0.5, 0.8$. **Right:** $\text{SO}(3)$ drawn as a schematic curved surface (a three-sphere-quotient cartoon is acceptable and standard). R_0 sampled Haar-uniformly, R_1 the crystal orientation, the geodesic between them, and the body-frame target u_t^{rot} drawn at the same three times — *drawn with identical length at every t* , since that is the content of Proposition 5.1. The visual point of the figure is that the right-hand arrows do not shrink as $t \rightarrow 1$. This replaces roughly 250 words and is the figure the reader should remember when reaching Chapter 5.

Figure 4.1: Conditional probability paths on the two factors of $\mathcal{M}$. Left: the straight-line translational path. Right: the geodesic rotational path, with the conditional target drawn at three times.

Chapter 5

A Testable Rotation–Translation Asymmetry

The experiments of Chapter 7 report a pronounced difference between the two factors of $\mathcal{M}$: the translational channel carries signal while the rotational channel, at the budgets examined so far, does not. This chapter derives a candidate mechanism from the construction of §3.5. It is presented as a *hypothesis with explicit falsifiers*, not as an established conclusion; §5.3 states what would refute it and §7.2 carries out the test.

5.1 The geodesic target does not depend on time

Proposition 5.1 (Time-invariance of the conditional rotational target). *Let $R_t = R_0 \text{Exp}(t \text{Log}(R_0^\top R_1))$ be the geodesic interpolant of (3.17), with Log taken on the principal branch. Then for all $t \in [0, 1)$,*

$$u_t^{\text{rot}} = \frac{\text{Log}(R_t^\top R_1)}{1 - t} = \text{Log}(R_0^\top R_1), \quad (5.1)$$

which is independent of t . In particular $\|u_t^{\text{rot}}\|$ is constant along each conditional trajectory and equals the full initial geodesic distance.

Proof. Write $\Delta = R_0^\top R_1$ and $\xi = \text{Log } \Delta$. Then $R_t^\top R_1 = \text{Exp}(-t\xi) R_0^\top R_1 = \text{Exp}(-t\xi) \text{Exp}(\xi)$. Both factors are exponentials of multiples of the same Lie algebra element and therefore commute, so the product is $\text{Exp}((1 - t)\xi)$. Taking logarithms gives $\text{Log}(R_t^\top R_1) = (1 - t)\xi$, and dividing by $1 - t$ yields (5.1). $\square$

The identity is not an artefact of the analysis: it is present in the implementation, where the training target is computed as $\text{Log}(R_0^\top R_1)$ directly and the code comments justify this as a numerically stabler form of $\text{Log}(R_t^\top R_1)/(1 - t)$ near $t = 1$. The property was exploited as a convenience before it was recognised as a mechanism.

The corresponding translational statement is the same, and equally standard: the target $x_1 - x_0$ of (3.16) is likewise constant in t . The asymmetry therefore cannot come from time-invariance alone. It comes from what the model can *infer* at each time, and that is a property

of the source.

5.2 Shrinkage of the conditional mean

By (3.4), the minimiser of the training objective is not the conditional target but its expectation given the current state:

$$v_{\theta}^*(t, R_t) = \mathbb{E}[\text{Log}(R_0^\top R_1) \mid R_t, c]. \quad (5.2)$$

Consider t near zero. Then $R_t \approx R_0 \sim \mathcal{U}_{\text{SO}(3)}$, and by the bi-invariance of the Haar measure R_0 is independent of R_1 ; the state supplies no information about the target beyond what c already provides. The expectation in (5.2) is then an average of rotation vectors over the whole conditional law of R_1 given c , and averaging vectors contracts:

$$\|\mathbb{E}[\xi]\| \leq \mathbb{E}[\|\xi\|], \quad (5.3)$$

with equality only if the directions of ξ coincide almost surely. For a conditional law that is at all diffuse in orientation the inequality is strict, and the optimal prediction is systematically shorter than the targets it is regressing on. A field of deficient magnitude integrates, through (3.19), to a rotation smaller than required; the sample ends near R_0 , that is, near Haar.

Nothing in this argument is special to flow matching *per se* — it is the ordinary behaviour of an L^2 regression under weak conditioning. What is specific is the combination: because the target is time-invariant, there is no regime in which the model is asked for a small, easy correction. It is asked for the entire answer at every t , including at $t = 0$ where the answer is least identifiable.

5.2.1 Why translation escapes, and what diffusion does differently

The translational source is $\mathcal{N}(0, I)$ in pocket-centred coordinates, so x_0 is concentrated near the origin and the data x_1 are concentrated near the pocket as well. The conditional mean $\mathbb{E}[x_1 - x_0 \mid x_t]$ is therefore a useful, non-degenerate answer even at $t = 0$: “move towards the pocket centre” is approximately correct for every complex. The rotational source admits no analogue. The Haar measure on $\text{SO}(3)$ is invariant under the full group action, so it has no distinguished mean orientation — every point is as central as every other. There is no direction in which the model can usefully hedge.

Diffusion differs in a way that is easy to state once Proposition 5.1 is in hand. The score $\nabla \log p_\tau$ of a heavily noised marginal is small in magnitude and smooth, so at high noise the model is asked for a small correction, and the difficult, sharply conditioned regime arises only at low noise. The noise schedule thereby acts as an implicit curriculum. The geodesic flow-matching construction discards that curriculum, and it is exactly the curriculum that the rotational channel, lacking an informative prior, appears to require.

5.3 Predictions and falsification criteria

The mechanism above is only useful if it can be wrong. It implies:

- P1 Magnitude deficit.** The ratio $\|v_\theta^{\text{rot}}\|/\|u_t^{\text{rot}}\|$ is below unity, most severely at small t . *Refuted if* the ratio is at or above unity across t .
- P2 Directional collapse at small t .** The cosine between predicted and target rotational fields is near zero for $t \rightarrow 0$ and increases with t . *Refuted if* early alignment is already substantial.
- P3 Persistence under compute.** Because the deficit is attributed to identifiability rather than to under-training, it should not disappear merely with a longer schedule. *Refuted if* rotational enrichment rises to the diffusion baseline given more compute — in which case the correct reading is slower convergence, not a structural obstruction.
- P4 Responsiveness to the source.** Replacing $\mathcal{U}_{\text{SO}(3)}$ by a concentrated, context-dependent source should improve the rotational channel specifically, and should not be required for translation. *Refuted if* an informative source leaves rotation unchanged.

P1 and P2 are measurable on an existing checkpoint and are reported in §7.3. P3 requires the matched-compute trajectory of §7.2. P4 is the subject of §7.6 and is the only one of the four that is also a candidate remedy.

[CHECK WITH SUPERVISOR] Whether to state P1–P4 as a formal pre-registration, with the reading rules fixed before the longer runs complete. The experiments were designed before the results existed, so the claim would be honest, and it strengthens the negative-result case considerably.

Chapter 6

Experimental Design

6.1 Data, training, and the matched-budget protocol

Both arms train on the PDBbind general set [9] and are evaluated on the PoseBusters benchmark [1], 209 complexes after the filtering inherited from the parent codebase. Splits, fragmentation, conformer generation and featurisation are shared, so the two arms consume byte-identical inputs.

Matched budget here means identical wall-clock allocation on identical hardware, identical batch size, identical optimiser and identical annealing schedule — not identical loss values or identical numbers of epochs, which are outcomes rather than controls. Table 6.1 records the configurations.

Table 6.1: Training configurations. The 12 h pair is the matched comparison available at the time of writing; the long pair is the primary experiment.

Run	Arm	Wall-clock	Hardware	Batch	Status
Early pair	SigmaFlow	12 h	1×L40S	8	complete
Early pair	SigmaDock	12 h	1×L40S	8	complete
Long pair	SigmaFlow	72 h	1×L40S	TBD	[RESULT PENDING] queued
Long pair	SigmaDock	72 h	1×L40S	TBD	[RESULT PENDING] queued
Source screen	SigmaFlow-informative	6 h	1×L40S	TBD	[RESULT PENDING] gated
Source screen	SigmaFlow-Haar control	6 h	1×L40S	TBD	[RESULT PENDING] gated

For calibration against the literature: SigmaDock’s published results were obtained with roughly 384 GPU-hours, whereas the early pair here used about 12 each. This thesis therefore operates at a small fraction of the compute behind the published numbers, and absolute accuracies are correspondingly low for *both* arms. The comparison remains valid because the budget is matched; the benchmark comparison is not attempted.

6.2 Trajectory snapshots are not independently annealed endpoints

The long runs checkpoint at fixed wall-clock intervals, and it is tempting to read the resulting sequence as performance “at 12 h, at 24 h, and so on”. That reading is wrong and the distinction is maintained throughout Chapter 7.

The learning rate follows a cosine schedule whose length is set by the total step budget. A snapshot taken part-way through a 72 h schedule is therefore a model *mid-anneal*, at a learning rate far above its terminal value, and it is not the model one would obtain by training for that shorter budget with a schedule annealed to it. Two distinct objects follow:

- a **training trajectory** — metrics against optimisation step for a *fixed* schedule, with both arms sharing that schedule. This is a controlled comparison and it answers whether the rotational deficit narrows as training proceeds;
- **annealed endpoints** — separately scheduled runs, each having completed its own anneal. The 12 h pair and the 72 h endpoint are of this kind.

Trajectories are plotted as curves; annealed endpoints are plotted as isolated markers and never joined to them. No claim of the form “performance scales with compute as ...” is made, since establishing that would require a separate fully annealed run per budget.

6.3 Metrics, nulls, and statistical treatment

Primary quantities are chosen for low sampling variance, so that a trajectory is readable rather than noisy:

- median per-fragment rotation error, against the Haar reference (2.7), and the enrichment below 30° ;
- median fragment centroid displacement;
- median RMSD, raw and after optimal superposition;
- validation loss, separately for the rotational and translational terms;
- the diagnostic ratios of §5.3: predicted-to-target magnitude ratio, and predicted-target cosine stratified by t .

Threshold success rates — the fraction below $2\text{ \AA}$ — are reported in tables with confidence intervals but are *not* plotted over training. At the rates observed so far and $n = 209$, their standard error is of the same order as any plausible change, so a curve would display sampling noise.

All paired comparisons are computed per complex, with bootstrap confidence intervals over complexes. Where several step counts or variants are compared against a common reference, the multiplicity is stated and no individual comparison is claimed significant on the strength

of an uncorrected interval. Stochasticity across sampling seeds is reported separately from stochasticity across complexes, since the two have been observed to be of comparable size and conflating them would overstate precision.

Chapter 7

Results

7.1 Early matched-budget behaviour

Table 7.1 reports the matched 12 h pair. Both models are far from the accuracies reported in the literature, which is expected at this budget (§6); the comparison of interest is between the columns.

Table 7.1: Matched 12 h comparison on PoseBusters (209 complexes, 10 sampling seeds per complex). “Single seed” averages the per-seed rate; “best of 10” is the oracle over seeds and is an upper bound on what any ranking function could achieve.

	SigmaFlow	SigmaDock
RMSD < 2 Å, single seed	4.4%	9.8%
RMSD < 2 Å, best of 10	29.2%	45.9%
Median RMSD, single seed	4.78 Å	4.48 Å
Median RMSD, best of 10	2.62 Å	2.10 Å

SigmaDock leads on every entry. This is the honest starting point of the empirical account, and it is stated before any interpretation.

Two features of the table deserve comment. First, the gap between the single-seed and best-of-10 rows is very large for both arms: for SigmaFlow, 4.4% against 29.2%. Most of the achievable accuracy at this budget is present in the sample set and lost at selection — a fact that motivates §7.7. Second, the two arms differ by roughly a factor of two throughout, rather than by a constant offset, which suggests a difference in kind rather than in degree. §7.3 localises it.

7.2 Training trajectory at matched compute

RESERVED – Training trajectory, 72 h matched pair

Experiment. Two runs, SigmaFlow and SigmaDock, identical schedule annealed to the full budget, checkpointed at 6, 12, 18, 24, 36, 48, 60, 72 h. Cheap evaluation on a fixed 60-complex subset at every snapshot; full 209-complex, 3-seed evaluation at 24, 48 and 72 h.

Expected output. Figure 7.1 (four panels) and a table of annealed endpoints with bootstrap intervals.

Metrics. Median fragment rotation error against the Haar reference; enrichment below 30°; median

centroid displacement; median RMSD raw and aligned; rotational and translational validation loss separately.

Question answered. Prediction P3 of §5.3: does the rotational deficit narrow, persist, or vanish with additional training?

Interpretation rules, fixed in advance. (i) If SigmaFlow’s rotational enrichment rises towards SigmaDock’s, P3 is refuted and the correct reading is *slower convergence*; the mechanism of Chapter 5 then explains the rate rather than a limit, and §8.2 says so. (ii) If SigmaFlow’s enrichment stays flat at the Haar level while SigmaDock’s rises, P3 is supported and the deficit is attributed to the construction. (iii) If both stay near Haar, no claim about the mechanism is licensed at this budget and the honest reading is that neither model has learned orientation; §8.5 must then lead with that. Reserved: ≈ 700 words.

FIGURE PLACEHOLDER

Four panels sharing an x -axis of optimisation step (secondary axis: wall-clock hours). Both arms on every panel, distinguished by colour. **(a)** median fragment rotation error, with the Haar reference 132.3° as a horizontal dashed line; **(b)** fraction of fragments with rotation error below 30° , with the Haar rate ($\approx 2\%$) as a dashed line; **(c)** median fragment centroid displacement; **(d)** median RMSD, raw and after superposition. Bands: bootstrap intervals over complexes. Annealed endpoints (12 h pair, 72 h endpoint) as filled markers *not* joined to the curves. Panel (b) is the decisive one: it shows whether SigmaFlow ever separates from the null.

Figure 7.1: Training trajectory at matched compute. Curves are snapshots from a single annealed schedule; isolated markers are independently annealed endpoints (§6.2).

7.3 Translation versus rotation

The factor-wise decomposition of the 12 h models localises the gap unambiguously. Table 7.2 reports per-fragment rotation error against the Haar null of §2.4.

Table 7.2: Per-fragment rotational accuracy at 12 h (711 fragments over 209 complexes, seed 0), against a uniformly random orientation.

	SigmaFlow	SigmaDock	Haar (random)
Median rotation error	138.2°	123.8°	132.3°
Fraction below 30°	2.1%	5.8%	$\approx 2\%$
Fraction below 60°	8.4%	15.8%	$\approx 10\%$

SigmaFlow’s rotational channel is not merely weaker than SigmaDock’s: it is not distinguishable from guessing. Its enrichment below 30° matches the random rate, whereas SigmaDock achieves roughly a threefold enrichment at identical architecture, data and budget.

The translational channel behaves quite differently. The fraction of fragments placed within $2 \text{ \AA}$ is 22.6% for SigmaFlow against 29.3% for SigmaDock, and within $1 \text{ \AA}$ 8.2% against 9.2% — a modest deficit, but signal in both cases rather than its absence. The difference between the two factors is therefore qualitative, not a matter of degree, which is what Chapter 5 sets out to explain.

FIGURE PLACEHOLDER

Three overlaid kernel density estimates on $[0, 180^\circ]$: SigmaFlow, SigmaDock, and the analytic Haar density. Vertical lines at the three medians. Inset: zoom on $[0, 60^\circ]$ where the enrichment differences live. The point of the figure is immediate legibility — the SigmaFlow density should be seen to sit on top of the Haar density while SigmaDock’s is shifted left. Replaces roughly 300 words. Requires the per-fragment error arrays, which exist for the 12 h models; regenerate for the 72 h models.

Figure 7.2: Distribution of per-fragment rotation error at 12 h, against the random-orientation null.

RESERVED – Mechanism diagnostics over training

Experiment. Evaluate predictions P1 and P2 of §5.3 at each checkpoint of the long runs: the ratio $\|v_\theta^{\text{rot}}\|/\|u_t^{\text{rot}}\|$ and the cosine between predicted and target rotational fields, both stratified by t .

Prior evidence. On the 12 h model the magnitude ratio is approximately 0.47 and the cosine at $t < 0.1$ approximately 0.025, both consistent with the shrinkage argument of §5.2.

Question answered. Whether shrinkage is an early-training artefact that resolves, or a stable property of the optimum.

Interpretation. A magnitude ratio rising towards 1 with training refutes P1 and points to under-training; a ratio stable well below 1 supports the identifiability reading. Reserved: ≈ 300 words within §7.3 (total for the section: ≈ 600).

7.4 Integration efficiency

The conditional paths of §3.5 are straight in the translational factor and geodesic in the rotational one, so the marginal field they induce should be nearly straight and few integration steps should suffice. Table 7.3 tests this over a forty-fold range.

Table 7.3: SigmaFlow accuracy against the number of Euler steps (209 complexes, 3 seeds). The inherited default is 25.

Steps	Median RMSD ($\pm$ SD over seeds)	$< 2 \text{ \AA}$	$< 5 \text{ \AA}$	Aligned RMSD
5	4.61 ± 0.06	3.7%	54.9%	2.11
10	4.78 ± 0.22	3.5%	53.0%	1.97
15	4.88 ± 0.11	4.0%	51.8%	1.95
25	4.76 ± 0.19	4.1%	53.3%	2.01
50	5.08 ± 0.05	4.0%	49.1%	1.91
100	5.17 ± 0.18	4.8%	48.8%	1.99
200	4.93 ± 0.24	3.7%	50.7%	1.99

The curve is flat. Paired per complex against the default of 25 steps, the largest effect is at 5 steps ($-0.199 \text{ \AA}$, 95% CI $[-0.391, -0.007]$); the upper limit is essentially at zero, and with six comparisons against a common reference this does not survive any correction for multiplicity. The defensible statement is therefore that *no step count between 5 and 200 is demonstrably better than any other*. For scale, a single complex varies across three sampling seeds with a standard deviation of $1.36 \text{ \AA}$, whereas its seed-averaged range across all seven step counts is $1.66 \text{ \AA}$ — less than would be expected from pure noise over seven draws.

Two readings follow, and both are needed. Positively, SigmaFlow reaches at 5 steps what it reaches at 25, so sampling can be made five times cheaper at no measurable cost — the signature predicted for near-straight conditional paths, and the clearest advantage of the substitution observed so far. Negatively, integration resolution is thereby *excluded* as an explanation for the accuracy gap: the model integrates its field faithfully; the field points in the wrong direction.

The claim licensed by this table is “SigmaFlow does not need the inherited 25 steps”, not “SigmaFlow needs fewer steps than SigmaDock”. The latter requires the matching sweep on the diffusion arm.

RESERVED – Diffusion-arm step sweep

Experiment. The same seven step counts and three seeds on the SigmaDock arm, run strictly serially so that wall-clock is interpretable. **Expected output.** A second series in Figure 7.3, and a two-sided efficiency claim. **Question answered.** Whether the few-step property is specific to flow matching or shared. **Interpretation.** If SigmaDock also tolerates 5 steps, the efficiency argument for flow matching collapses and must be withdrawn; if it degrades, the advantage is established. Reserved: ≈ 150 words.

FIGURE PLACEHOLDER

Median RMSD (left axis) against integration steps on a log x -axis, 5 to 200. Bands from three seeds. Two series once the diffusion sweep exists. A second panel may show wall-clock per pose against steps, but *only* from serially executed jobs — the existing array was run with five concurrent tasks and its timings are contaminated by resource contention.

Figure 7.3: Accuracy against number of function evaluations, both arms.

7.5 Diagnostics and ablations

7.5.1 The frame correction and its signature

The defect of §4.2 was identified analytically, but its behavioural signature confirms that the evaluation is sensitive to a known real intervention. We measure a *locality gap*: the difference between the rotation error of a fragment and that of its neighbours, negative when a model places neighbouring fragments more coherently than chance.

Table 7.4: Locality gap before and after the frame correction ($n \approx 133$ complexes, bootstrap intervals over complexes).

Run	Locality gap	95% CI	Fraction < 0
SigmaFlow 6 h, uncorrected	+1.1°	[−4.2, +6.6]	53%
SigmaFlow 6 h, corrected	−12.8°	[−18.3, −7.2]	68%
SigmaFlow 12 h, corrected	−15.0°	[−21.6, −8.1]	69%
SigmaDock 12 h	−20.4°	[−26.0, −14.5]	71%

The uncorrected model is indistinguishable from no local coherence; the correction moves it decisively. This licenses the use of the same metric elsewhere.

7.5.2 What does not explain the gap

Three candidate explanations were tested and none survives.

- **Integration resolution** — excluded by §7.4.
- **Molecular symmetry.** Symmetry-equivalent atom assignments could in principle inflate apparent rotation errors. Measured directly, they do not account for the observed distribution.
- **Loss weighting and objective variants.** Three single-change ablations off the same baseline — a time-weighted flow loss, a rotational loss in data space rather than tangent space, and an anchor-atom distance term — all returned a locality gap of -12.8° , statistically indistinguishable from the unmodified baseline. Table 7.5 summarises.

Table 7.5: Single-change ablations, each branched from the same baseline.

Variant	Single change	Locality gap
Baseline (corrected)	—	-15.0°
(a) time weighting	$w(t)\ v_\theta - u\ ^2$	-12.8°
(b) rotational loss in data space	loss on R rather than $\text{Log } R$	-12.8°
(c) anchor-atom distance	auxiliary distance term	-12.8°

That three independent modifications of the objective produce no change is itself evidence: it is consistent with a model whose rotational channel has no signal to reweight. One caveat is recorded — the reconstructed fragmentation differs between runs for a substantial minority of complexes, so bond-level comparisons across these variants are less reliable than the fragment-level rotational statistics used here.

7.6 Flow-specific source distribution

§3.7 established that p_0 is free under flow matching and fixed under diffusion, and §5.2 argued that the uninformative nature of $\mathcal{U}_{\text{SO}(3)}$ is what starves the rotational channel. Prediction P4 follows: concentrating the rotational source should improve rotation specifically.

Two constraints shape the construction. It must be *sampleable* and tractable, for which a wrapped Gaussian on $\text{SO}(3)$ about a centre R_c suffices,

$$R_0 = R_c \text{Exp}(\hat{\xi}), \quad \xi \sim \mathcal{N}(0, \sigma^2 I_3), \quad (7.1)$$

with right-multiplication chosen to match the body-frame convention of §3.4.1. And it must not leak: R_c may depend only on the receptor and the ligand graph, never on the crystallographic pose. This is not a formality — an earlier variant in this project was found to place conformer fragments at crystal centroids, which supplies the answer through the source, and the constraint is therefore enforced explicitly and tested by a control in which (7.1) reduces to $\mathcal{U}_{\text{SO}(3)}$ and must reproduce the baseline bit-for-bit.

RESERVED – Informative-source experiment

Gate. A CPU-only audit measures whether receptor pocket geometry carries orientational information at all, by comparing the rotational distance under a candidate heuristic centre against the Haar reference of 132.3° . *Pre-registered rule:* if the median distance is not below the Haar reference by more than its bootstrap interval, the experiment is not run and this section reports the audit as a negative result in ≈ 350 words.

Experiment, if gated through. A 6 h screen with (7.1) against a bit-identical 6 h Haar control from the same code path, the two differing only in p_0 .

Expected output. Table 7.6 completed; Figure 7.4.

Metrics. Median rotation error and enrichment below 30° (primary, since P4 predicts a rotation-specific effect); centroid displacement and RMSD (secondary, expected approximately unchanged).

Interpretation. A rotation-specific improvement supports the mechanism causally rather than correlationally and is the strongest available evidence for Chapter 5. An improvement in *both* factors would instead suggest a generic optimisation benefit and would weaken the specific claim. No improvement refutes P4. Reserved: ≈ 600 words.

Table 7.6: Effect of the source distribution on the rotational channel, at matched budget.

Source p_0 on $\text{SO}(3)$	Median rotation error	$< 30^\circ$	Median RMSD
Haar (baseline)	TBD	TBD	TBD
Wrapped Gaussian, $\sigma = \text{TBD}$	TBD	TBD	TBD
Haar reference (random)	132.3°	$\approx 2\%$	—

FIGURE PLACEHOLDER

Two panels. **(a)** Distribution of the initial geodesic distance $\|\text{Log}(R_0^\top R_1)\|$ under the Haar source and under (7.1), with the Haar mean 126.5° marked. By Proposition 5.1 this is also the distribution of the target magnitude at every t , which is the point of the panel. **(b)** Rotation-error densities for the two trained variants, as in Figure 7.2, with the Haar null. Contingent on the gate.

Figure 7.4: Source distributions on $\text{SO}(3)$ and their effect on the conditional target.

7.7 Confidence and pose ranking

Table 7.1 shows that selection, not generation, is where most achievable accuracy is lost at this budget: 4.4% against a 29.2% oracle. A ranking function that recovered even a third of that difference would exceed any architectural effect measured in this thesis.

RESERVED – Confidence head

Experiment. A small head on the $\ell = 0$ (rotation-invariant) channel of the existing backbone, trained in a second stage to predict $\mathbb{P}(\text{RMSD} < 2 \text{ \AA})$ for a generated pose. Two-stage rather than joint, because during generative training the network sees only x_t , which at $t = 1$ is exactly the crystal pose; there are no poses of varying quality to learn from.

Expected output. Table 7.7; Figure 7.5.

Metrics. AUROC, AUPRC, Brier score, expected calibration error; Spearman correlation against RMSD; and the operational quantity, top-1-of- K accuracy against random-of- K and the oracle.

Question answered. Whether the inherited backbone’s invariant features carry pose-quality information at all.

Scope note. The comparison against SigmaDock’s published heuristic ranker requires an external scoring binary that is not currently available in the compute environment; if it remains unavailable, this section reports calibration and top-1-of- K only, and says so. *This extension is not flow-specific* — an equivalent head could be attached to SigmaDock — and must be presented as a property of the system, not of flow matching. Reserved: ≈ 300 words; drops to a Future Work paragraph if the head is not trained.

Table 7.7: Confidence-head calibration and ranking performance.

	AUROC	Brier	Spearman vs. RMSD	Top-1-of-10 < 2 Å
Random selection	0.50	—	0	TBD
Confidence head	TBD	TBD	TBD	TBD
Oracle (best of 10)	—	—	—	29.2%

FIGURE PLACEHOLDER

Three panels: (a) reliability diagram with the diagonal and per-bin counts; (b) top- k accuracy against k , for random selection, the confidence head, and the oracle, showing how much of the oracle gap is recovered; (c) mean predicted confidence against integration time t along the sampling trajectory, testing whether confidence becomes informative before $t = 1$ and could in principle support early termination. Contingent.

Figure 7.5: Confidence-head behaviour.

Chapter 8

Discussion

8.1 What the substitution changed

Replacing score-based diffusion by Riemannian flow matching, with everything else held fixed, changed three things that can be stated independently of the outcome of the longer runs.

It changed *what the network is asked to predict*. A score has units of inverse state and must be rescaled by noise-level-dependent factors before it can move a pose; a velocity is already in the units of the state. The entire scalar apparatus that performed that conversion was deleted rather than adapted (§4.1), which is a genuine simplification of the model’s interface, not merely a reorganisation.

It changed *the cost of sampling*. The conditional paths are straight or geodesic by construction, and §7.4 confirms that accuracy is flat from 5 to 200 integration steps — a fivefold reduction against the inherited default at no measurable cost. Whether this is an advantage *over diffusion* awaits the matching sweep on the other arm, and the claim is stated in the weaker form until then.

And it changed *what is a free parameter*. Under diffusion the source is determined by the forward process; under flow matching it is a modelling choice (§3.7). This is the degree of freedom that Chapter 5 identifies as consequential and §7.6 exploits.

8.2 Verdict on the asymmetry hypothesis

RESERVED – Verdict, written against the final evidence

This section states which of P1–P4 (§5.3) survived. It is deliberately left unwritten: writing it now would mean committing to a conclusion before the test. The four anticipated readings and the shape each would take:

(A) Rotation catches up with compute. P3 refuted. The mechanism is then a statement about *convergence rate*, not a structural limit: the time-invariant target makes the rotational problem harder to optimise early, but not unlearnable. The efficiency result of §7.4 becomes the headline, and the thesis argues that flow matching is competitive above a compute threshold that should be quantified.

(B) The deficit persists. P3 supported. The thesis argues that the geodesic path with a Haar source is

structurally deficient for *conditional* generation on $\text{SO}(3)$, and that the diffusion noise schedule functions as an implicit curriculum that the flow-matching construction discards. §7.6 then becomes the causal test of that claim.

(C) Minimal remains worse, informative source repairs it. P3 and P4 both supported, and the strongest available form of the argument: the deficiency is diagnosed *and* located in a specific, replaceable component, with the repair following from the mathematics rather than from search. Source design is then the thesis’s main methodological claim.

(D) Accuracy parity, advantage elsewhere. The asymmetry was a transient; the substitution is accuracy-neutral but cheaper to sample and, if §7.7 succeeds, better equipped for selection. The thesis then argues on cost and calibration rather than accuracy, with the explicit caveat that confidence estimation is not flow-specific.

Reserved: ≈ 400 words.

8.3 Is the comparison genuinely controlled?

The strength of any claim here rests on the substitution being isolated, so the residual differences must be stated rather than assumed away.

In favour: the backbone source is byte-identical between arms, verified by hash; featurisation, splits, fragmentation, optimiser, schedule and loss weights are shared; both arms were trained on the same hardware for the same wall-clock time with the same batch size; and evaluation used one pipeline for both.

Against: the two arms execute in separate software environments with different interpreter versions, so a difference attributable to library behaviour cannot be excluded outright. The diffusion arm is invoked with its own rotational-score configuration, which is required for it to function but is nonetheless a difference. And the reconstructed fragmentation differs between runs for a substantial minority of complexes, which is why the analysis relies on fragment-level rotational statistics rather than bond-level geometry.

None of these plausibly produces a threefold difference in rotational enrichment, but they are recorded so the reader can weigh them.

8.4 Conceptual differences independent of accuracy

Some consequences of the substitution do not depend on which model scores higher. Training is simulation-free and requires no noise schedule to tune, so one hyperparameter family disappears. Sampling is a deterministic map from source sample to pose, which makes the sample set a function of $\{x_0^{(k)}\}$ alone and simplifies reasoning about diversity. The learned object is a well-defined ODE, so exact likelihood is available in principle through the instantaneous change-of-variables formula — with the caveat that the divergence on $\text{SO}(3)$ requires care that this thesis has not verified, and which is therefore listed as future work rather than claimed. And the source is explicit, which is what makes §7.6 possible at all.

8.5 Limitations

Compute. All runs here are far below the budget behind SigmaDock’s published results. Absolute accuracies are correspondingly low for both arms, and no statement about benchmark-level performance is made. The comparison is internally matched, which is the claim that is actually defended.

A single training seed per arm. Run-to-run variability in training is not characterised, so a difference of the size seen between the two arms cannot be formally separated from seed variation in training. Sampling-seed variability *is* characterised and is reported throughout.

Evaluation. PoseBusters is a single benchmark; RMSD-centric metrics are known to be imperfect proxies for binding-mode correctness; and the torsional-fingerprint metric explored as a complement proved essentially uncorrelated with RMSD and failed to compute for a substantial fraction of molecules, so it is not reported.

Inherited design. The rigid-fragment approximation, the fragmentation algorithm, and the single-head Newton–Euler reduction are all inherited. In particular, the last of these means translation and rotation are two functionals of one output, and a genuinely separate rotational head might behave differently — an intervention deliberately excluded here because it is not flow-specific and would have confounded the comparison.

Scope of the mechanism. The argument of §5.2 is a statement about L^2 regression under weak conditioning. It predicts that small fragments, whose rotational readout is most poorly conditioned, should rotate worst. That prediction is *not* visible in the data (135.8° for small fragments against 141.7° for large ones in SigmaFlow). With a rotational channel already at chance there is little dynamic range in which to detect such a trend, so the sub-prediction is recorded as unconfirmed rather than as supporting evidence.

8.6 Future work

Three directions follow directly. A separate equivariant rotational head would test whether the two channels compete through their shared readout; this is the intervention with the largest expected effect and the least flow specificity, and that combination should be stated plainly. Exact ODE likelihood is attractive for ranking, but requires first validating the divergence computation on the manifold against an analytically known test field. And source design generalises: the argument of Chapter 5 applies to any conditional generative model on a homogeneous space whose natural invariant measure is uninformative, which includes several problems in structural biology beyond docking.

Chapter 9

Conclusion

This thesis replaced the score-based diffusion mechanism of a rigid-fragment protein–ligand docking model with Riemannian flow matching on $(\text{SE}(3))^N$, holding the equivariant backbone, the data pipeline and the computational budget fixed so that the substitution itself could be measured.

Three results are established independently of the longer training runs. The substitution simplifies the model’s interface, replacing a score that requires noise-dependent rescaling with a velocity that does not. It permits a fivefold reduction in integration steps at no measurable cost in accuracy, the signature expected of near-straight conditional paths. And it exposes an asymmetry between the factors of the state space: at matched budget the translational channel carries signal while the rotational channel is statistically indistinguishable from uniform guessing.

The principal theoretical contribution is a candidate explanation for that asymmetry. The geodesic conditional target on $\text{SO}(3)$ is independent of time (Proposition 5.1), so flow matching demands the complete rotational answer at every t , including $t = 0$ where the state is uninformative; the L^2 -optimal response is then a contracting conditional mean. Translation escapes this because its source is centred on the binding pocket and therefore informative, whereas the Haar measure has no distinguished orientation to hedge towards. Diffusion’s noise schedule supplies, incidentally, the graded curriculum that the flow-matching construction discards.

[RESULT PENDING] The remaining two paragraphs state the verdict on predictions P1–P4 and on the informative-source intervention, and are written only once Chapter 7 is complete. The four candidate shapes are set out in §8.2. Reserved: ≈ 150 words.

Beyond docking, the analysis suggests a general point. When a conditional generative model is built on a homogeneous space, the invariant measure is the obvious source distribution and is often the only one considered — but invariance is precisely the property that makes it uninformative. Flow matching, unlike diffusion, leaves that choice open. Whether exploiting it repairs the rotational channel here is the question the remaining experiments are designed to answer.

Bibliography

- [1] Martin Buttenschön, Garrett M. Morris, and Charlotte M. Deane. Posebusters: Ai-based docking methods fail to generate physically valid poses or generalise to novel sequences. *Chemical Science*, 15:3130–3139, 2024.
- [2] Ricky T. Q. Chen and Yaron Lipman. Flow matching on general geometries. In *International Conference on Learning Representations*, 2024.
- [3] Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, and David Duvenaud. Neural ordinary differential equations. In *Advances in Neural Information Processing Systems*, 2018.
- [4] Gabriele Corso, Hannes Stärk, Bowen Jing, Regina Barzilay, and Tommi Jaakkola. Diffdock: Diffusion steps, twists, and turns for molecular docking. In *International Conference on Learning Representations*, 2023.
- [5] Wolfgang Kabsch. A solution for the best rotation to relate two sets of vectors. *Acta Crystallographica Section A*, 32(5):922–923, 1976.
- [6] Adam Leach, Sebastian M. Schmon, Matteo T. Degiacomi, and Chris G. Willcocks. Denoising diffusion probabilistic models on $so(3)$ for rotational alignment. In *ICLR Workshop on Geometrical and Topological Representation Learning*, 2022.
- [7] Yi-Lun Liao, Brandon Wood, Abhishek Das, and Tess Smidt. Equiformerv2: Improved equivariant transformer for scaling to higher-degree representations. In *International Conference on Learning Representations*, 2024.
- [8] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023.
- [9] Zhihai Liu, Minyi Su, Li Han, Jie Liu, Qifan Yang, Yan Li, and Renxiao Wang. Forging the basis for developing protein–ligand interaction scoring functions. *Accounts of Chemical Research*, 50(2):302–309, 2017.
- [10] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019.
- [11] Jerrold E. Marsden and Tudor S. Ratiu. *Introduction to Mechanics and Symmetry*. Springer, 2nd edition, 1999.

- [12] Saro Passaro and C. Lawrence Zitnick. Reducing $so(3)$ convolutions to $so(2)$ for efficient equivariant gnns. In *International Conference on Machine Learning*, 2023.
- [13] A. Prat et al. Sigmadock: a diffusion model for protein–ligand docking. 2026. Bibliographic details to be completed from the publisher record.
- [14] Rodrigo Quiroga and Marcos A. Villarreal. Vinardo: A scoring function based on autodock vina improves scoring, docking, and virtual screening. *PLoS ONE*, 11(5):e0155183, 2016.
- [15] Yang Song and Stefano Ermon. Generative modeling by estimating gradients of the data distribution. In *Advances in Neural Information Processing Systems*, 2019.
- [16] Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2021.
- [17] Pascal Vincent. A connection between score matching and denoising autoencoders. *Neural Computation*, 23(7):1661–1674, 2011.
- [18] Jason Yim, Brian L. Trippe, Valentin De Bortoli, Emile Mathieu, Arnaud Doucet, Regina Barzilay, and Tommi Jaakkola. $Se(3)$ diffusion model with application to protein backbone generation. In *International Conference on Machine Learning*, 2023.

Appendix A

Equivariance and the Inherited Backbone

§2.3.1 states only the interface between the backbone and the generative mechanism, on the grounds that the backbone is inherited unchanged and is not a contribution of this thesis. This appendix supplies the representation-theoretic background for readers who want it.

[DRAFT] Compress from the corresponding chapters of the project's theory document. Target ≈ 900 words. Cover, in this order: group actions and the formal definitions of invariance and equivariance; irreducible representations of $SO(3)$ and the Wigner- D matrices; spherical harmonics as the basis in which steerable features are expressed; tensor products and Clebsch–Gordan coefficients as the mechanism by which features of different degree interact; the $SO(2)$ reduction of Passaro and Zitnick [12] that makes EquiformerV2 tractable [7]; and the architecture's hyperparameters and parameter count. No derivations — statements, one worked shape example, and citations. Nothing here is needed to follow the main argument, and the appendix should say so in its first sentence.

Appendix B

Diffusion Background

[DRAFT] Compress to ≈ 600 words. Stochastic differential equations and the Euler–Maruyama scheme; denoising score matching [17]; the variance-preserving and variance-exploding families [16]; Brownian motion on a Riemannian manifold and the heat kernel on $SO(3)$, i.e. IGSO(3) [6]; and the probability-flow ODE with the affine relation between score and velocity that connects the two formulations. Statements and citations, no derivations: §2.3 and §3.6 carry everything the argument uses.

Appendix C

Derivations

[DRAFT] Full versions of the three results stated compactly in the main text: the marginalisation theorem (Proposition 3.1) with the Riemannian case treated explicitly rather than by remark; the adjoint-transport identity (3.15) with the derivation of both trivialisations from the group action; and the Haar expected-angle computation (2.7), which is a short integral against the $SO(3)$ angle density and is worth doing in full because every rotational claim in Chapter 7 is measured against it.

Appendix D

Numerical Verification

[DRAFT] Tabulate the geometric verification suite that supports Chapter 4. Contents already exist as executable checks and should be reported as a table of quantity, expected value, observed value: the frame convention identified from the imported code rather than from comments; the Euler step of length $1 - t$ reaching R_1 (observed 5×10^{-4} degrees) against the incorrect left-multiplication (observed $\approx 179.5^\circ$); the adjoint identity to machine precision; accuracy of the logarithmic map before and after the clamp correction of §4.3; and the Haar null verified both analytically and over 20,000 samples. Include the note on catastrophic cancellation in the chord-form angle computation, which is a genuine numerical trap and cost real debugging time.

Appendix E

Extended Results

[DRAFT] Material displaced from Chapter 7 for space: per-seed breakdowns of Table 7.1; the full step sweep with all paired intervals; seed-variance analysis; the PoseBusters check-by-check breakdown; and the complete experiment registry, including variants that were specified and rejected on mathematical grounds without being run. The last of these is worth including — rejecting an experiment for a stated reason is part of the work, and the registry records why optimal-transport coupling cannot apply when the target is a single fixed pose per complex.

Appendix F

Implementation Reference

[DRAFT] Short. Numerical safeguards and their justification (clamps, epsilons, the float64 islands in otherwise float32 code); masking, batching and scatter conventions; and a table mapping each mathematical object in Chapter 3 to the module that implements it. This is the only place in the thesis where file and function names belong; the main text refers to equations, not to code.